PetrT_17122024_5451

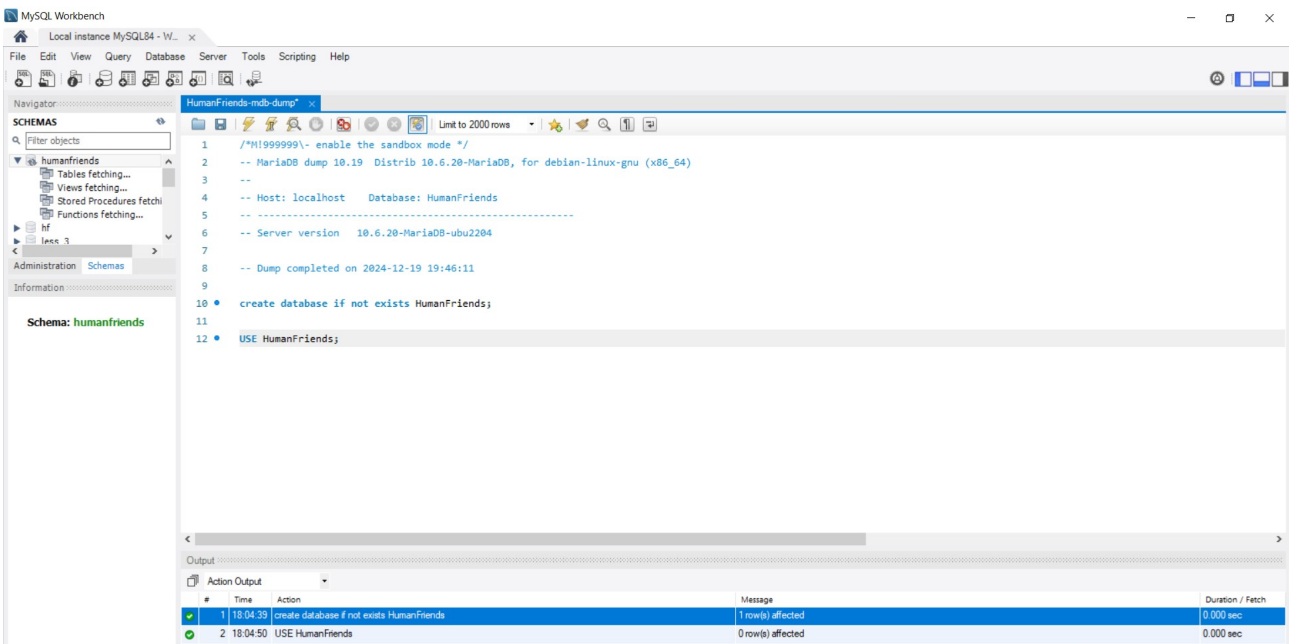
По заданию необходимо использование СУБД, установленной на linux-машине.
Подключился по ssh к своей ВМ. Создал БД «HumanFriends».

```
gb-linux# mysql
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 34
Server version: 10.6.20-MariaDB-ubu2204 mariadb.org binary distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> create database if not exists HumanFriends;
Query OK, 0 rows affected, 1 warning (0,000 sec)

MariaDB [(none)]> show databases;
+--------------------+
| Database           |
+--------------------+
| HumanFriends       |
| information_schema |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
5 rows in set (0,001 sec)

MariaDB [(none)]> use HumanFriends;
Database changed
MariaDB [HumanFriends]>
```

Для проверки создал логический backup моей БД в общей папке.

```
gb-linux# mysqldump -u root -p HumanFriends > HumanFriends-mdb-dump.sql
Enter password:
gb-linux# tree
.
├── animals
│   └── HumanFriends.txt
├── HumanFriends-mdb-dump.sql
├── PackAnimals.txt
└── Pets.txt

1 directory, 4 files
```

Вроде все работает.
Далее, для удобства, действия с таблицами буду производить в MysqlWorkbrench



Результат:

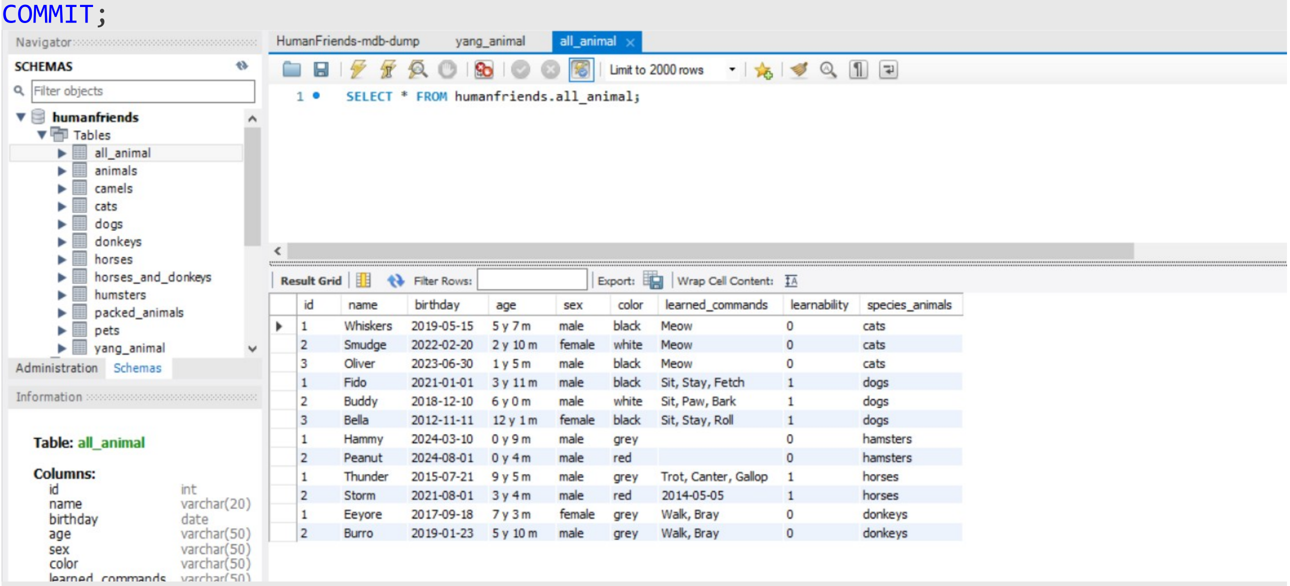
```sql
CREATE DATABASE IF NOT EXISTS HumanFriends;

USE HumanFriends;

-- Создать таблицы, соответствующие иерархии из вашей диаграммы классов.
CREATE TABLE animals(
  animals_id INT NOT NULL,
  type_animals VARCHAR(50) NOT NULL PRIMARY KEY
);

INSERT INTO animals(animals_id, type_animals)
VALUES
  (1, 'Pets'),
  (2, 'Packed Animals');

CREATE TABLE pets(
  pets_id INT NOT NULL,
  species_pets VARCHAR(50) NOT NULL PRIMARY KEY,
  type_animals VARCHAR(50) NOT NULL,
  FOREIGN KEY (type_animals) REFERENCES animals (type_animals) ON DELETE CASCADE ON
UPDATE CASCADE
);

INSERT INTO pets(pets_id, species_pets, type_animals)
VALUES
  (1, 'cats', 'Pets'),
  (2, 'dogs', 'Pets'),
  (3, 'hamsters', 'Pets');

CREATE TABLE packed_animals(
  packed_animals_id INT NOT NULL,
  species_packedanimals VARCHAR (50) NOT NULL PRIMARY KEY,
  type_animals VARCHAR(50) NOT NULL,
  FOREIGN KEY (type_animals) REFERENCES animals (type_animals) ON DELETE CASCADE ON
UPDATE CASCADE
);

INSERT INTO packed_animals(packed_animals_id, species_packedanimals, type_animals)
VALUES
  (1, 'horses', 'Packed Animals'),
  (2, 'camels', 'Packed Animals'),
  (3, 'donkeys', 'Packed Animals');

CREATE TABLE cats(
  cats_id INT AUTO_INCREMENT NOT NULL PRIMARY KEY,
  name VARCHAR(20) NOT NULL,
  birthday DATE NOT NULL,
  age VARCHAR(50) NOT NULL,
  sex VARCHAR(50) NOT NULL,
```

```sql
  color VARCHAR(50) NOT NULL,
  learned_commands VARCHAR(50),
  learnability BOOLEAN NOT NULL,
  species_pets VARCHAR(50) NOT NULL,
  Foreign KEY (species_pets) REFERENCES pets (species_pets) ON DELETE CASCADE ON
UPDATE CASCADE
);

CREATE TABLE dogs(
  dogs_id INT AUTO_INCREMENT NOT NULL PRIMARY KEY,
  name VARCHAR(20) NOT NULL,
  birthday DATE NOT NULL,
  age VARCHAR(50) NOT NULL,
  sex VARCHAR(50) NOT NULL,
  color VARCHAR(50) NOT NULL,
  learned_commands VARCHAR(50),
  learnability BOOLEAN NOT NULL,
  species_pets VARCHAR(50) NOT NULL,
  Foreign KEY (species_pets) REFERENCES pets (species_pets) ON DELETE CASCADE ON
UPDATE CASCADE
);

CREATE TABLE humsters(
  humsters_id INT AUTO_INCREMENT NOT NULL PRIMARY KEY,
  name VARCHAR(20) NOT NULL,
  birthday DATE NOT NULL,
  age VARCHAR(50) NOT NULL,
  sex VARCHAR(50) NOT NULL,
  color VARCHAR(50) NOT NULL,
  learned_commands VARCHAR(50),
  learnability BOOLEAN NOT NULL,
  species_pets VARCHAR(50) NOT NULL,
  Foreign KEY (species_pets) REFERENCES pets (species_pets) ON DELETE CASCADE ON
UPDATE CASCADE
);

CREATE TABLE horses(
  horses_id INT AUTO_INCREMENT NOT NULL PRIMARY KEY,
  name VARCHAR(20) NOT NULL,
  birthday DATE NOT NULL,
  age VARCHAR(50) NOT NULL,
  sex VARCHAR(50) NOT NULL,
  color VARCHAR(50) NOT NULL,
  learned_commands VARCHAR(50),
  learnability BOOLEAN NOT NULL,
  species_packedanimals VARCHAR(50) NOT NULL,
  Foreign KEY (species_packedanimals) REFERENCES packed_animals
(species_packedanimals) ON DELETE CASCADE ON UPDATE CASCADE
);

CREATE TABLE camels(
```

```
  camels_id INT AUTO_INCREMENT NOT NULL PRIMARY KEY,
  name VARCHAR(20) NOT NULL,
  birthday DATE NOT NULL,
  age VARCHAR(50) NOT NULL,
  sex VARCHAR(50) NOT NULL,
  color VARCHAR(50) NOT NULL,
  learned_commands VARCHAR(50),
  learnability BOOLEAN NOT NULL,
  species_packedanimals VARCHAR(50) NOT NULL,
  Foreign KEY (species_packedanimals) REFERENCES packed_animals
(species_packedanimals) ON DELETE CASCADE ON UPDATE CASCADE
);

CREATE TABLE donkeys(
  donkeys_id INT AUTO_INCREMENT NOT NULL PRIMARY KEY,
  name VARCHAR(20) NOT NULL,
  birthday DATE NOT NULL,
  age VARCHAR(50) NOT NULL,
  sex VARCHAR(50) NOT NULL,
  color VARCHAR(50) NOT NULL,
  learned_commands VARCHAR(50),
  learnability BOOLEAN NOT NULL,
  species_packedanimals VARCHAR(50) NOT NULL,
  Foreign KEY (species_packedanimals) REFERENCES packed_animals
(species_packedanimals) ON DELETE CASCADE ON UPDATE CASCADE
);

-- Заполнить таблицы данными о животных, их командах и датами рождения.
INSERT INTO cats(name, birthday, age, sex, color, learned_commands, learnability,
species_pets)
VALUES
  ('Whiskers', '2019-05-15',
  CONCAT(TIMESTAMPDIFF(YEAR, birthday, CURDATE()), ' y ', TIMESTAMPDIFF(MONTH,
birthday, CURDATE()) % 12, ' m '), 'male', 'black', 'Meow', 0, 'cats'),
  ('Smudge', '2022-02-20',
  CONCAT(TIMESTAMPDIFF(YEAR, birthday, CURDATE()), ' y ', TIMESTAMPDIFF(MONTH,
birthday, CURDATE()) % 12, ' m '), 'female', 'white', 'Meow', 0, 'cats'),
  ('Oliver', '2023-06-30',
  CONCAT(TIMESTAMPDIFF(YEAR, birthday, CURDATE()), ' y ', TIMESTAMPDIFF(MONTH,
birthday, CURDATE()) % 12, ' m '), 'male', 'black', 'Meow', 0, 'cats');

INSERT INTO dogs(name, birthday, age, sex, color, learned_commands, learnability,
species_pets)
VALUES
  ('Fido', '2021-01-01',
  CONCAT(TIMESTAMPDIFF(YEAR, birthday, CURDATE()), ' y ', TIMESTAMPDIFF(MONTH,
birthday, CURDATE()) % 12, ' m '), 'male', 'black', 'Sit, Stay, Fetch', 1, 'dogs'),
  ('Buddy', '2018-12-10',
  CONCAT(TIMESTAMPDIFF(YEAR, birthday, CURDATE()), ' y ', TIMESTAMPDIFF(MONTH,
birthday, CURDATE()) % 12, ' m '), 'male', 'white', 'Sit, Paw, Bark', 1, 'dogs'),
  ('Bella', '2012-11-11',
```

```sql
  CONCAT(TIMESTAMPDIFF(YEAR, birthday, CURDATE()), ' y ', TIMESTAMPDIFF(MONTH,
birthday, CURDATE()) % 12, ' m '), 'female', 'black', 'Sit, Stay, Roll', 1,
'dogs');

INSERT INTO humsters(name, birthday, age, sex, color, learned_commands,
learnability, species_pets)
VALUES
  ('Hammy', '2024-03-10',
  CONCAT(TIMESTAMPDIFF(YEAR, birthday, CURDATE()), ' y ', TIMESTAMPDIFF(MONTH,
birthday, CURDATE()) % 12, ' m '), 'male', 'grey', '', 0, 'hamsters'),
  ('Peanut', '2024-08-01',
  CONCAT(TIMESTAMPDIFF(YEAR, birthday, CURDATE()), ' y ', TIMESTAMPDIFF(MONTH,
birthday, CURDATE()) % 12, ' m '), 'male', 'red', '', 0, 'hamsters');

INSERT INTO horses(name, birthday, age, sex, color, learned_commands, learnability,
species_packedanimals)
VALUES
  ('Thunder', '2015-07-21',
  CONCAT(TIMESTAMPDIFF(YEAR, birthday, CURDATE()), ' y ', TIMESTAMPDIFF(MONTH,
birthday, CURDATE()) % 12, ' m '), 'male', 'grey', 'Trot, Canter, Gallop', 1,
'horses'),
  ('Storm', '2021-08-01',
  CONCAT(TIMESTAMPDIFF(YEAR, birthday, CURDATE()), ' y ', TIMESTAMPDIFF(MONTH,
birthday, CURDATE()) % 12, ' m '), 'male', 'red', 'Trot, Canter', 1, 'horses');

INSERT INTO camels(name, birthday, age, sex, color, learned_commands, learnability,
species_packedanimals)
VALUES
  ('Sandy', '2016-11-03',
  CONCAT(TIMESTAMPDIFF(YEAR, birthday, CURDATE()), ' y ', TIMESTAMPDIFF(MONTH,
birthday, CURDATE()) % 12, ' m '), 'male', 'orange', 'Walk, Carry Load', 0,
'camels'),
  ('Dune', '2018-12-12',
  CONCAT(TIMESTAMPDIFF(YEAR, birthday, CURDATE()), ' y ', TIMESTAMPDIFF(MONTH,
birthday, CURDATE()) % 12, ' m '), 'male', 'broun', 'Walk, Sit', 0, 'camels');

INSERT INTO donkeys(name, birthday, age, sex, color, learned_commands,
learnability, species_packedanimals)
VALUES
  ('Eeyore', '2017-09-18',
  CONCAT(TIMESTAMPDIFF(YEAR, birthday, CURDATE()), ' y ', TIMESTAMPDIFF(MONTH,
birthday, CURDATE()) % 12, ' m '), 'female', 'grey', 'Walk, Bray', 0, 'donkeys'),
  ('Burro', '2019-01-23',
  CONCAT(TIMESTAMPDIFF(YEAR, birthday, CURDATE()), ' y ', TIMESTAMPDIFF(MONTH,
birthday, CURDATE()) % 12, ' m '), 'male', 'grey', 'Walk, Bray', 0, 'donkeys');

-- Удалить записи о верблюдах и объединить таблицы лошадей и ослов.
DELETE FROM camels;

CREATE TABLE horses_and_donkeys
  SELECT * FROM horses
```

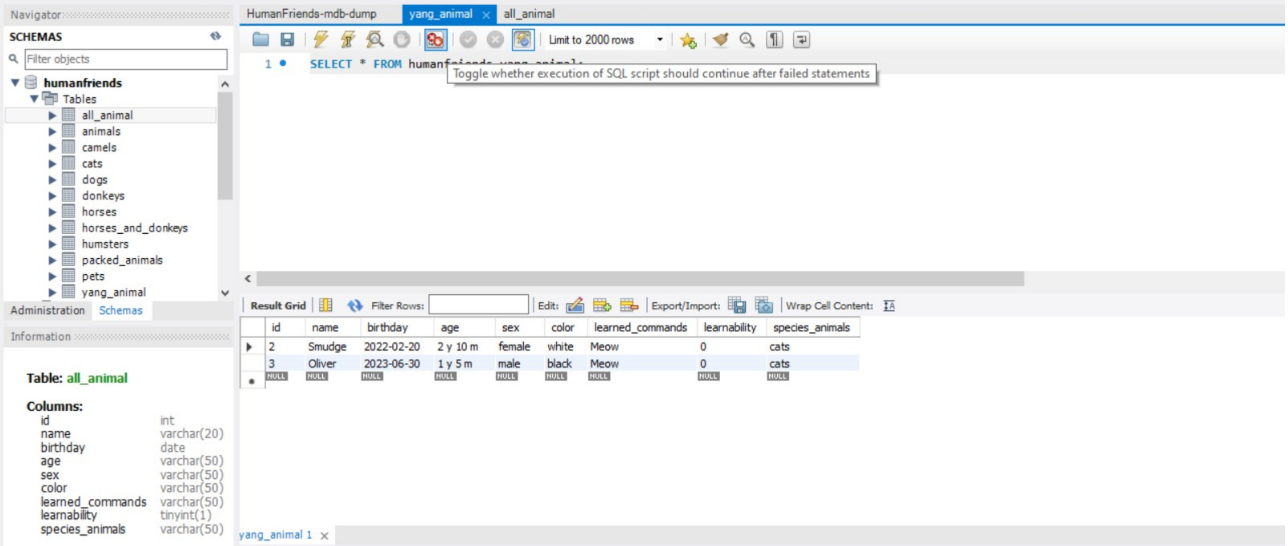
```sql
  UNION
  SELECT * FROM donkeys;

-- Объединить все созданные таблицы в одну, сохраняя информацию о принадлежности к исходным таблицам (изменил последовательность - считаю что так проще).
CREATE TABLE all_animal(
  id INT,
  name VARCHAR(20) NOT NULL,
  birthday DATE NOT NULL,
  age VARCHAR(50) NOT NULL,
  sex VARCHAR(50) NOT NULL,
  color VARCHAR(50) NOT NULL,
  learned_commands VARCHAR(50),
  learnability BOOLEAN NOT NULL,
  species_animals VARCHAR(50)
);

START TRANSACTION;
INSERT INTO all_animal(id, name, birthday, age, sex, color, learned_commands, learnability, species_animals) SELECT * FROM cats;
INSERT INTO all_animal(id, name, birthday, age, sex, color, learned_commands, learnability, species_animals) SELECT * FROM dogs;
INSERT INTO all_animal(id, name, birthday, age, sex, color, learned_commands, learnability, species_animals) SELECT * FROM humsters;
INSERT INTO all_animal(id, name, birthday, age, sex, color, learned_commands, learnability, species_animals) SELECT * FROM horses;
INSERT INTO all_animal(id, name, birthday, age, sex, color, learned_commands, learnability, species_animals) SELECT * FROM camels;
INSERT INTO all_animal(id, name, birthday, age, sex, color, learned_commands, learnability, species_animals) SELECT * FROM donkeys;
COMMIT;
```



id	name	birthday	age	sex	color	learned_commands	learnability	species_animals
1	Whiskers	2019-05-15	5 y 7 m	male	black	Meow	0	cats
2	Smudge	2022-02-20	2 y 10 m	female	white	Meow	0	cats
3	Oliver	2023-06-30	1 y 5 m	male	black	Meow	0	cats
1	Fido	2021-01-01	3 y 11 m	male	black	Sit, Stay, Fetch	1	dogs
2	Buddy	2018-12-10	6 y 0 m	male	white	Sit, Paw, Bark	1	dogs
3	Bella	2012-11-11	12 y 1 m	female	black	Sit, Stay, Roll	1	dogs
1	Hammy	2024-03-10	0 y 9 m	male	grey		0	hamsters
2	Peanut	2024-08-01	0 y 4 m	male	red		0	hamsters
1	Thunder	2015-07-21	9 y 5 m	male	grey	Trot, Canter, Gallop	1	horses
2	Storm	2021-08-01	3 y 4 m	male	red	2014-05-05	1	horses
1	Eeyore	2017-09-18	7 y 3 m	female	grey	Walk, Bray	0	donkeys
2	Burro	2019-01-23	5 y 10 m	male	grey	Walk, Bray	0	donkeys

```sql
-- Создать новую таблицу для животных в возрасте от 1 до 3 лет и вычислить их возраст с точностью до месяца.
CREATE TABLE yang_animal(
  id INT AUTO_INCREMENT NOT NULL PRIMARY KEY,
```

```
  name VARCHAR(20) NOT NULL,
  birthday DATE NOT NULL,
  age VARCHAR(50) NOT NULL,
  sex VARCHAR(50) NOT NULL,
  color VARCHAR(50) NOT NULL,
  learned_commands VARCHAR(50),
  learnability BOOLEAN NOT NULL,
  species_animals VARCHAR(50)
);

INSERT INTO yang_animal (id, name, birthday, age, sex, color, learned_commands,
learnability, species_animals)
  SELECT * FROM all_animal
    WHERE birthday BETWEEN ADDDATE(curdate(), INTERVAL -3 YEAR) AND
ADDDATE(CURDATE(), INTERVAL -1 YEAR);
```



Проверю работу в линуксе:

```
root@gb-linux: /home/gbpetr
    ->     color VARCHAR(50) NOT NULL,
    ->     learned_commands VARCHAR(50),
    ->     learnability BOOLEAN NOT NULL,
    ->     species_animals VARCHAR(50)
    -> );
ERROR 1050 (42S01): Table 'yang_animal' already exists
MariaDB [HumanFriends]>
MariaDB [HumanFriends]> INSERT INTO yang_animal (id, name, birthday, age, sex, color, learned_commands, learnability, species_animals)
    ->     SELECT * FROM all_animal
    ->         WHERE birthday BETWEEN ADDDATE(curdate(), INTERVAL -3 YEAR) AND ADDDATE(CURDATE(), INTERVAL -1 YEAR);
Query OK, 2 rows affected (0,003 sec)
Records: 2  Duplicates: 0  Warnings: 0

MariaDB [HumanFriends]> show databases;
+--------------------+
| Database           |
+--------------------+
| HumanFriends       |
| information_schema |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
5 rows in set (0,001 sec)

MariaDB [HumanFriends]> describe HumanFriends;
ERROR 1146 (42S02): Table 'HumanFriends.HumanFriends' doesn't exist
MariaDB [HumanFriends]> describe yang_animal;
+------------------+-------------+------+-----+---------+----------------+
| Field            | Type        | Null | Key | Default | Extra          |
+------------------+-------------+------+-----+---------+----------------+
| id               | int(11)     | NO   | PRI | NULL    | auto_increment |
| name             | varchar(20) | NO   |     | NULL    |                |
| birthday         | date        | NO   |     | NULL    |                |
| age              | varchar(50) | NO   |     | NULL    |                |
| sex              | varchar(50) | NO   |     | NULL    |                |
| color            | varchar(50) | NO   |     | NULL    |                |
| learned_commands | varchar(50) | YES  |     | NULL    |                |
| learnability     | tinyint(1)  | NO   |     | NULL    |                |
| species_animals  | varchar(50) | YES  |     | NULL    |                |
+------------------+-------------+------+-----+---------+----------------+
9 rows in set (0,002 sec)
```

```
MariaDB [HumanFriends]> describe all_animal;
+------------------+-------------+------+-----+---------+-------+
| Field            | Type        | Null | Key | Default | Extra |
+------------------+-------------+------+-----+---------+-------+
| id               | int(11)     | YES  |     | NULL    |       |
| name             | varchar(20) | NO   |     | NULL    |       |
| birthday         | date        | NO   |     | NULL    |       |
| age              | varchar(50) | NO   |     | NULL    |       |
| sex              | varchar(50) | NO   |     | NULL    |       |
| color            | varchar(50) | NO   |     | NULL    |       |
| learned_commands | varchar(50) | YES  |     | NULL    |       |
| learnability     | tinyint(1)  | NO   |     | NULL    |       |
| species_animals  | varchar(50) | YES  |     | NULL    |       |
+------------------+-------------+------+-----+---------+-------+
9 rows in set (0,001 sec)

MariaDB [HumanFriends]> SELECT * FROM all_animal;
+------+----------+------------+---------+--------+-------+-----------------------+--------------+-----------------+
| id   | name     | birthday   | age     | sex    | color | learned_commands      | learnability | species_animals |
+------+----------+------------+---------+--------+-------+-----------------------+--------------+-----------------+
|    1 | Whiskers | 2019-05-15 | 5 y 7 m | male   | black | Meow                  |            0 | cats            |
|    2 | Smudge   | 2022-02-20 | 2 y 10 m | female | white | Meow                 |            0 | cats            |
|    3 | Oliver   | 2023-06-30 | 1 y 5 m | male   | black | Meow                  |            0 | cats            |
|    1 | Fido     | 2021-01-01 | 3 y 11 m | male  | black | Sit, Stay, Fetch      |            1 | dogs            |
|    2 | Buddy    | 2018-12-10 | 6 y 0 m | male   | white | Sit, Paw, Bark        |            1 | dogs            |
|    3 | Bella    | 2012-11-11 | 12 y 1 m | female | black | Sit, Stay, Roll      |            1 | dogs            |
|    1 | Hammy    | 2024-03-10 | 0 y 9 m | male   | grey  |                       |            0 | hamsters        |
|    2 | Peanut   | 2024-08-01 | 0 y 4 m | male   | red   |                       |            0 | hamsters        |
|    1 | Thunder  | 2015-07-21 | 9 y 5 m | male   | grey  | Trot, Canter, Gallop  |            1 | horses          |
|    2 | Storm    | 2021-08-01 | 3 y 4 m | male   | red   | Trot, Canter          |            1 | horses          |
|    1 | Eeyore   | 2017-09-18 | 7 y 3 m | female | grey  | Walk, Bray            |            0 | donkeys         |
|    2 | Burro    | 2019-01-23 | 5 y 10 m | male  | grey  | Walk, Bray            |            0 | donkeys         |
+------+----------+------------+---------+--------+-------+-----------------------+--------------+-----------------+
12 rows in set (0,000 sec)

MariaDB [HumanFriends]> SELECT * FROM yang_animal;
+----+--------+------------+----------+--------+-------+------------------+--------------+-----------------+
| id | name   | birthday   | age      | sex    | color | learned_commands | learnability | species_animals |
+----+--------+------------+----------+--------+-------+------------------+--------------+-----------------+
|  2 | Smudge | 2022-02-20 | 2 y 10 m | female | white | Meow             |            0 | cats            |
|  3 | Oliver | 2023-06-30 | 1 y 5 m  | male   | black | Meow             |            0 | cats            |
+----+--------+------------+----------+--------+-------+------------------+--------------+-----------------+
```

все Ок!

Перезаписываю логический backup моей БД в общей папке

```
1 directory, 4 files
gb-linux# mysqldump -v -uroot -p -B HumanFriends >  HumanFriends-mdb-dump.sql
Enter password:
-- Connecting to localhost...
-- Retrieving table structure for table all_animal...
-- Sending SELECT query...
-- Retrieving rows...
-- Retrieving table structure for table animals...
-- Sending SELECT query...
-- Retrieving rows...
-- Retrieving table structure for table camels...
-- Sending SELECT query...
-- Retrieving rows...
-- Retrieving table structure for table cats...
-- Sending SELECT query...
-- Retrieving rows...
-- Retrieving table structure for table dogs...
-- Sending SELECT query...
-- Retrieving rows...
-- Retrieving table structure for table donkeys...
-- Sending SELECT query...
-- Retrieving rows...
-- Retrieving table structure for table horses...
-- Sending SELECT query...
-- Retrieving rows...
-- Retrieving table structure for table horses_and_donkeys...
-- Sending SELECT query...
-- Retrieving rows...
-- Retrieving table structure for table humsters...
-- Sending SELECT query...
-- Retrieving rows...
-- Retrieving table structure for table packed_animals...
-- Sending SELECT query...
-- Retrieving rows...
-- Retrieving table structure for table pets...
-- Sending SELECT query...
-- Retrieving rows...
-- Retrieving table structure for table yang_animal...
-- Sending SELECT query...
-- Retrieving rows...
-- Disconnecting from localhost...
gb-linux# |
```